

Africast-Time Series Analysis & Forecasting Using R

<https://workshop.f4sg.org/africast/>

Transforming / adjusting time series



Outline

- 1 Transforming time series
- 2 Adjusting time series
- 3 Time series decompositions
- 4 Multiple seasonality
- 5 The ABS stuff-up

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Variance stabilisation

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Mathematical transformations for stabilising variation

Square root	$w_t = \sqrt{y_t}$	$\downarrow$
Cube root	$w_t = \sqrt[3]{y_t}$	Increasing
Logarithm	$w_t = \log(y_t)$	strength

Variance stabilisation

If the data show different variation at different levels of the series, then a transformation can be useful.

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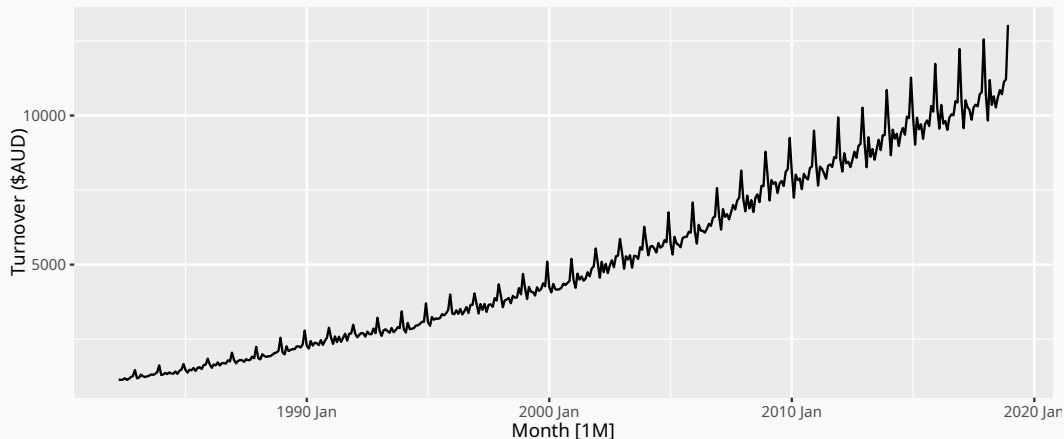
Mathematical transformations for stabilising variation

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Logarithms, in particular, are useful because they are more interpretable: changes in a log value are **relative (percent) changes**

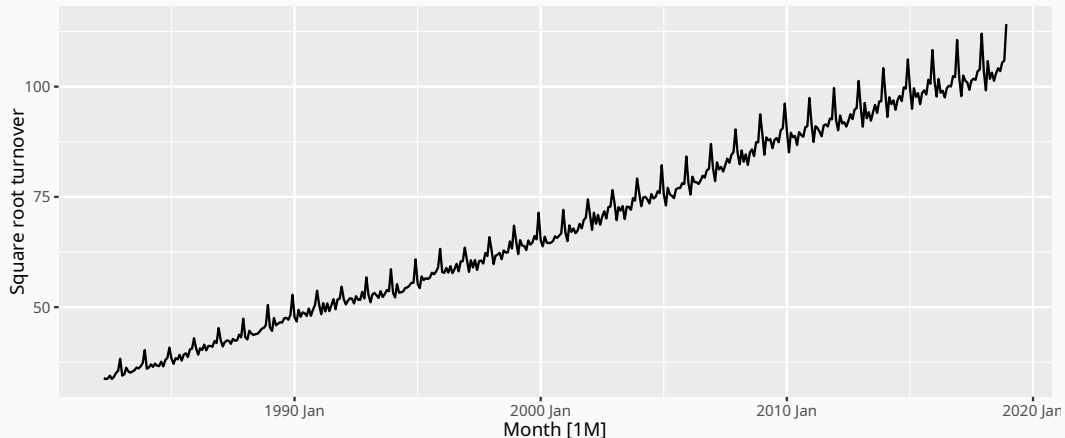
Variance stabilisation

```
food <- aus_retail |>  
  filter(Industry == "Food retailing") |>  
  summarise(Turnover = sum(Turnover))
```



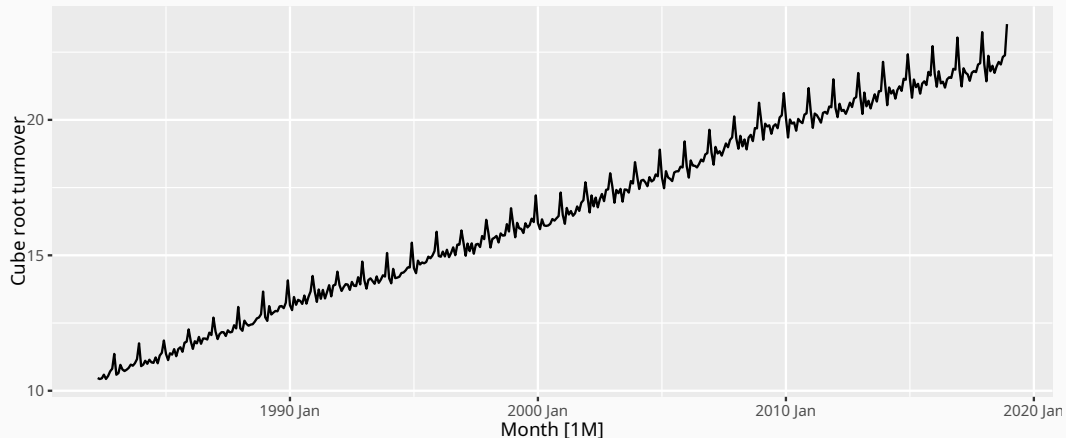
Variance stabilisation

```
food |> autoplot(sqrt(Turnover)) +  
  labs(y = "Square root turnover")
```



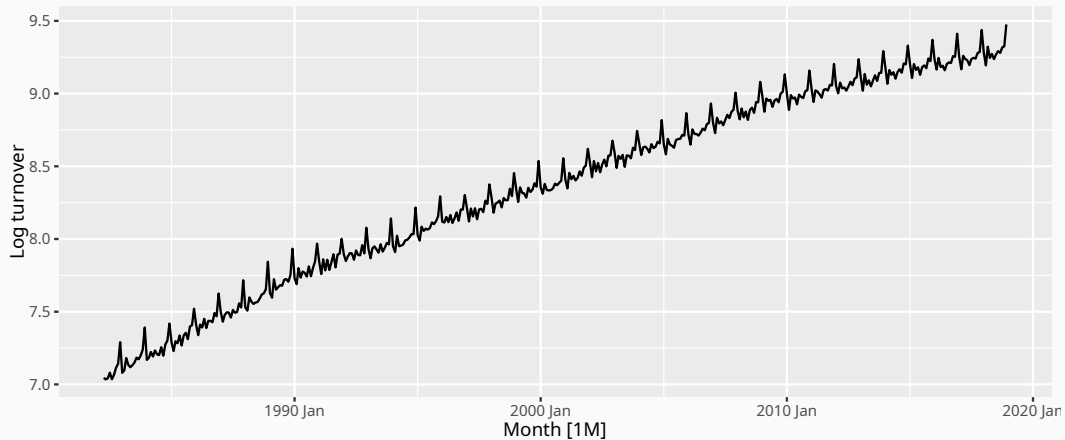
Variance stabilisation

```
food |> autoplot(Turnover^(1 / 3)) +  
  labs(y = "Cube root turnover")
```



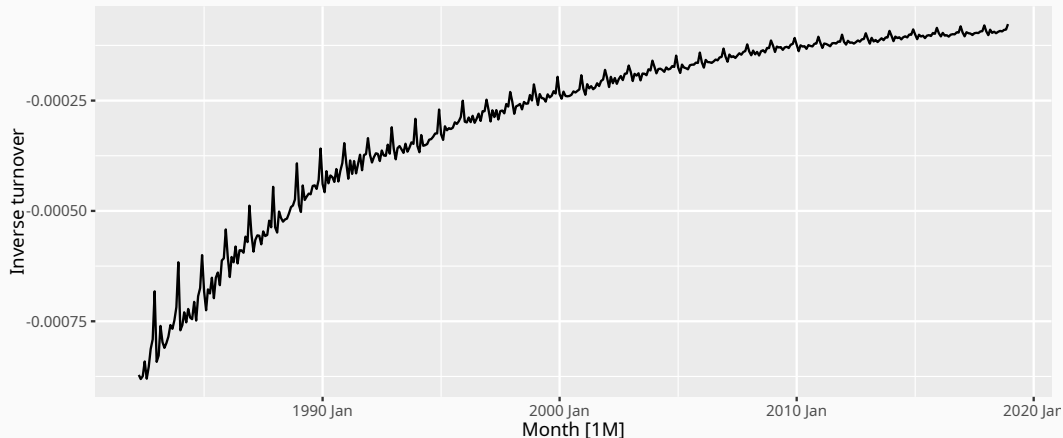
Variance stabilisation

```
food |> autoplot(log(Turnover)) +  
  labs(y = "Log turnover")
```



Variance stabilisation

```
food |> autoplot(-1 / Turnover) +  
  labs(y = "Inverse turnover")
```



Box-Cox transformations

Each of these transformations is close to a member of the family of **Box-Cox transformations**:

$$w_t = \begin{cases} \log(y_t), & \lambda = 0; \\ (\text{sign}(y_t)|y_t|^\lambda - 1)/\lambda, & \lambda \neq 0. \end{cases}$$

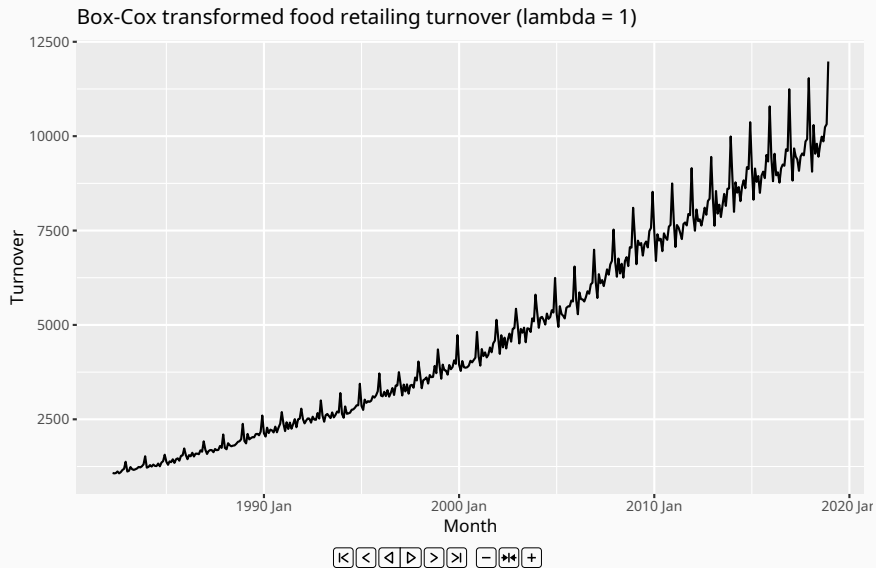
Box-Cox transformations

Each of these transformations is close to a member of the family of **Box-Cox transformations**:

$$w_t = \begin{cases} \log(y_t), & \lambda = 0; \\ (\text{sign}(y_t)|y_t|^\lambda - 1)/\lambda, & \lambda \neq 0. \end{cases}$$

- Actually the Bickel-Doksum transformation (allowing for $y_t < 0$)
- $\lambda = 1$: (No substantive transformation)
- $\lambda = \frac{1}{2}$: (Square root plus linear transformation)
- $\lambda = 0$: (Natural logarithm)
- $\lambda = -1$: (Inverse plus 1)

Box-Cox transformations



Box-Cox transformations

```
food |>  
  features(Turnover, features = guerrero)
```

```
# A tibble: 1 x 1  
  lambda_guerrero  
            <dbl>  
1             0.0895
```

Box-Cox transformations

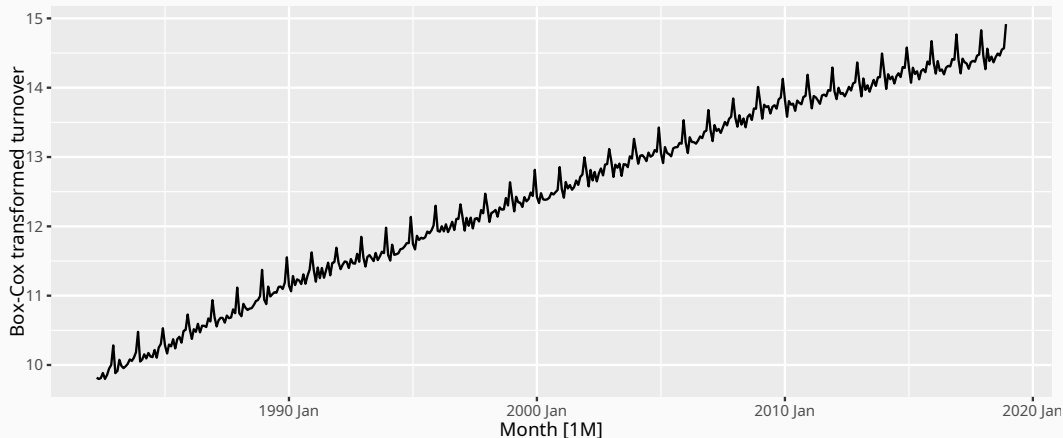
```
food |>  
  features(Turnover, features = guerrero)
```

```
# A tibble: 1 x 1  
  lambda_guerrero  
          <dbl>  
1             0.0895
```

- This attempts to balance the seasonal fluctuations and random variation across the series.
- Always check the results.
- A low value of λ can give extremely large prediction intervals.

Box-Cox transformations

```
food |> autoplot(box_cox(Turnover, 0.0895)) +  
  labs(y = "Box-Cox transformed turnover")
```



Transformations

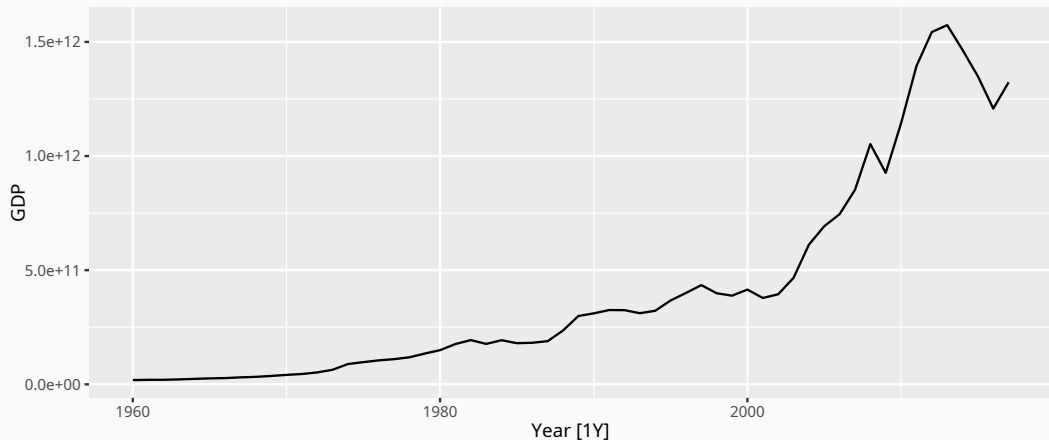
- Often no transformation needed.
- Simple transformations are easier to explain and work well enough.
- Transformations can have very large effect on PI.
- If some data are zero or negative, then use $\lambda > 0$.
- `log1p()` can also be useful for data with zeros.
- Choosing logs is a simple way to force forecasts to be positive
- Transformations must be reversed to obtain forecasts on the original scale. (Handled automatically by `fab1e`.)

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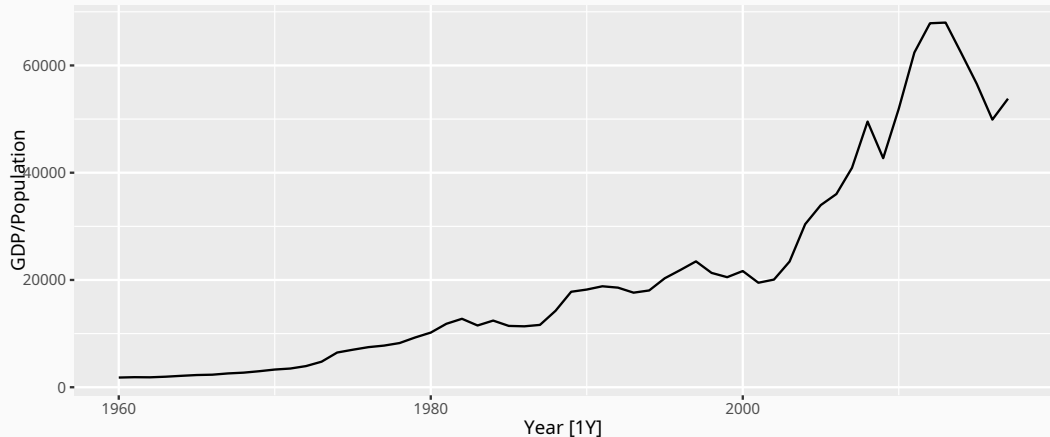
Per capita adjustments

```
global_economy |>  
  filter(Country == "Australia") |>  
  autoplot(GDP)
```



Per capita adjustments

```
global_economy |>  
  filter(Country == "Australia") |>  
  autoplot(GDP / Population)
```

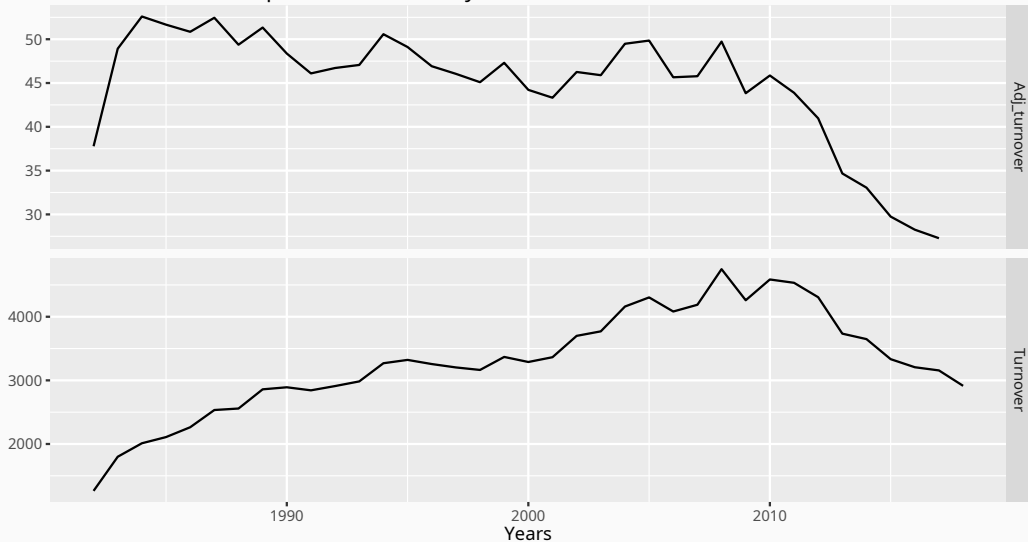


Inflation adjustments

```
print_retail <- aus_retail |>
  filter(Industry == "Newspaper and book retailing") |>
  group_by(Industry) |>
  index_by(Year = year(Month)) |>
  summarise(Turnover = sum(Turnover))
aus_economy <- filter(global_economy, Country == "Australia")
print_retail |>
  left_join(aus_economy, by = "Year") |>
  mutate(Adj_turnover = Turnover / CPI) |>
  pivot_longer(c(Turnover, Adj_turnover),
    names_to = "Type", values_to = "Turnover"
  ) |>
  ggplot(aes(x = Year, y = Turnover)) +
  geom_line() +
  facet_grid(vars(Type), scales = "free_y") +
  labs(x = "Years", y = NULL,
    title = "Turnover: Australian print media industry")
```

Inflation adjustments

Turnover: Australian print media industry



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Time series decomposition

Trend-Cycle aperiodic changes in level over time.

Seasonal (almost) periodic changes in level due to seasonal factors (e.g., the quarter of the year, the month, or day of the week).

Additive decomposition

$$y_t = S_t + T_t + R_t$$

where $y_t =$ data at period t

$T_t =$ trend-cycle component at period t

$S_t =$ seasonal component at period t

$R_t =$ remainder component at period t

STL decomposition

- STL: “Seasonal and Trend decomposition using Loess”
- Very versatile and robust.
- Seasonal component allowed to change over time, and rate of change controlled by user.
- Smoothness of trend-cycle also controlled by user.
- Optionally robust to outliers
- No trading day or calendar adjustments.
- Only additive.
- Take logs to get multiplicative decomposition.
- Use Box-Cox transformations to get other decompositions.

US Retail Employment

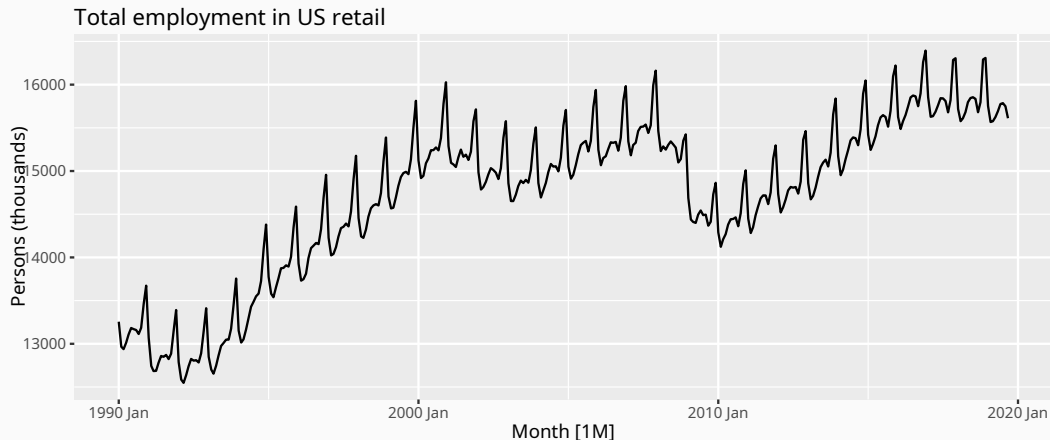
```
us_retail_employment <- us_employment |>
  filter(year(Month) >= 1990, Title == "Retail Trade") |>
  select(-Series_ID)
us_retail_employment
```

```
# A tsibble: 357 x 3 [1M]
```

	Month	Title	Employed
	<mth>	<chr>	<dbl>
1	1990 Jan	Retail Trade	13256.
2	1990 Feb	Retail Trade	12966.
3	1990 Mar	Retail Trade	12938.
4	1990 Apr	Retail Trade	13012.
5	1990 May	Retail Trade	13108.
6	1990 Jun	Retail Trade	13183.
7	1990 Jul	Retail Trade	13170.
8	1990 Aug	Retail Trade	13160.
9	1990 Sep	Retail Trade	13113.
10	1990 Oct	Retail Trade	13185.

US Retail Employment

```
us_retail_employment |>  
  autoplot(Employed) +  
  labs(y = "Persons (thousands)", title = "Total employment in US retail")
```



US Retail Employment

```
dcmp <- us_retail_employment |>  
  model(stl = STL(Employed))  
dcmp
```

```
# A mable: 1 x 1  
  stl  
  <model>  
1    <STL>
```

US Retail Employment

`components(dcmp)`

```
# A dable: 357 x 7 [1M]
```

```
# Key:      .model [1]
```

```
# :      Employed = trend + season_year + remainder
```

	.model	Month	Employed	trend	season_year	remainder	season_adjust
	<chr>	<mth>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	stl	1990 Jan	13256.	13288.	-33.0	0.836	13289.
2	stl	1990 Feb	12966.	13269.	-258.	-44.6	13224.
3	stl	1990 Mar	12938.	13250.	-290.	-22.1	13228.
4	stl	1990 Apr	13012.	13231.	-220.	1.05	13232.
5	stl	1990 May	13108.	13211.	-114.	11.3	13223.
6	stl	1990 Jun	13183.	13192.	-24.3	15.5	13207.
7	stl	1990 Jul	13170.	13172.	-23.2	21.6	13193.
8	stl	1990 Aug	13160.	13151.	-9.52	17.8	13169.
9	stl	1990 Sep	13113.	13131.	-39.5	22.0	13153.
10	stl	1990 Oct	13185.	13110.	61.6	13.2	13124.

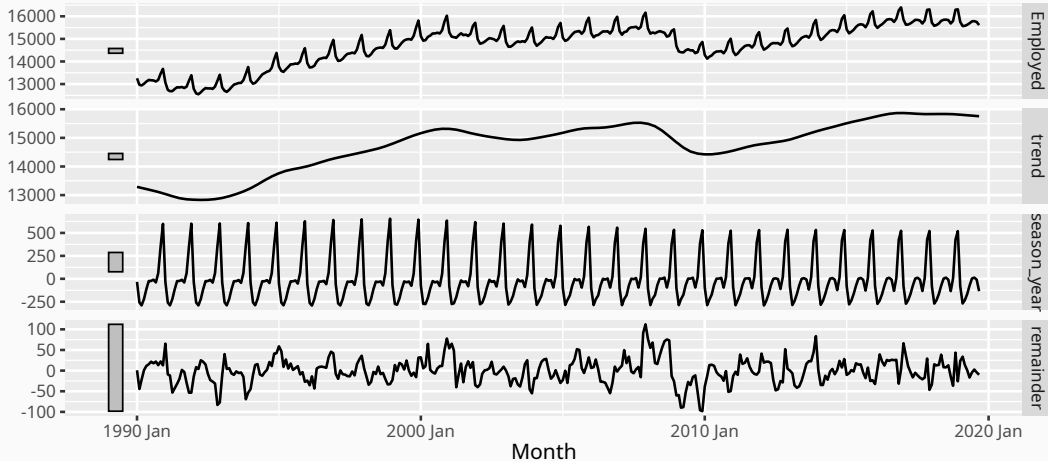
```
# : 347
```

US Retail Employment

```
components(dcmp) |> autoplot()
```

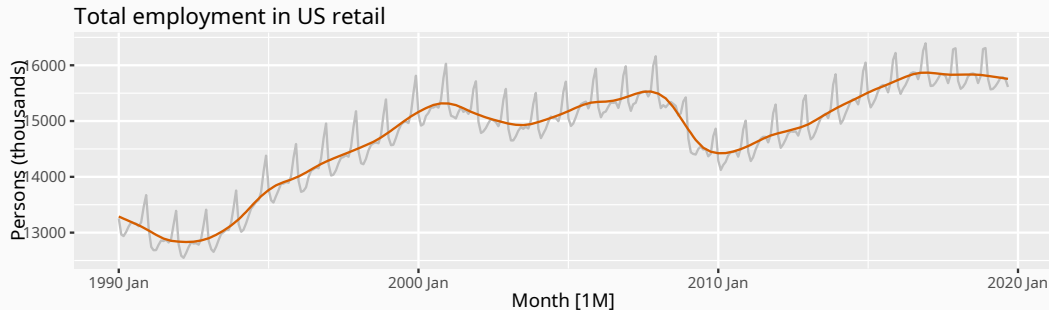
STL decomposition

Employed = trend + season_year + remainder



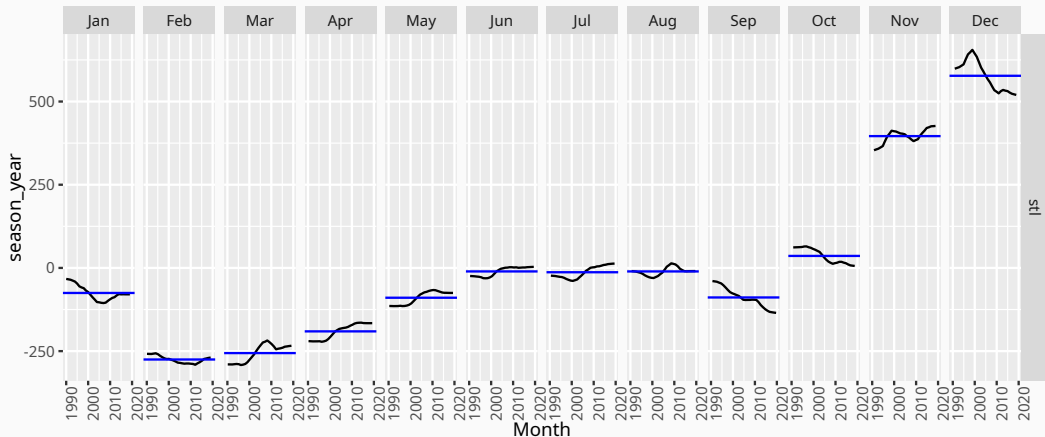
US Retail Employment

```
us_retail_employment |>  
  autoplot(Employed, color = "gray") +  
  autolayer(components(dcmp), trend, color = "#D55E00") +  
  labs(y = "Persons (thousands)", title = "Total employment in US retail")
```



US Retail Employment

```
components(dcmp) |> gg_subseries(season_year)
```



Seasonal adjustment

- Useful by-product of decomposition: an easy way to calculate seasonally adjusted data.
- Additive decomposition: seasonally adjusted data given by

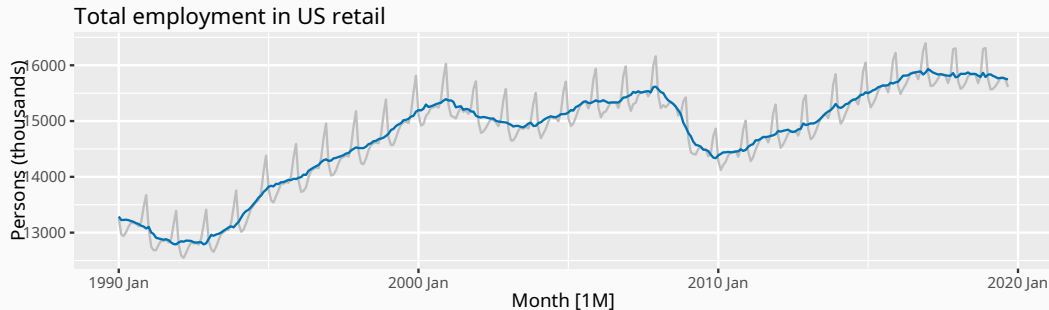
$$y_t - S_t = T_t + R_t$$

- Multiplicative decomposition: seasonally adjusted data given by

$$y_t / S_t = T_t \times R_t$$

US Retail Employment

```
us_retail_employment |>  
  autoplot(Employed, color = "gray") +  
  autolayer(components(dcmp), season_adjust, color = "#0072B2") +  
  labs(y = "Persons (thousands)", title = "Total employment in US retail")
```



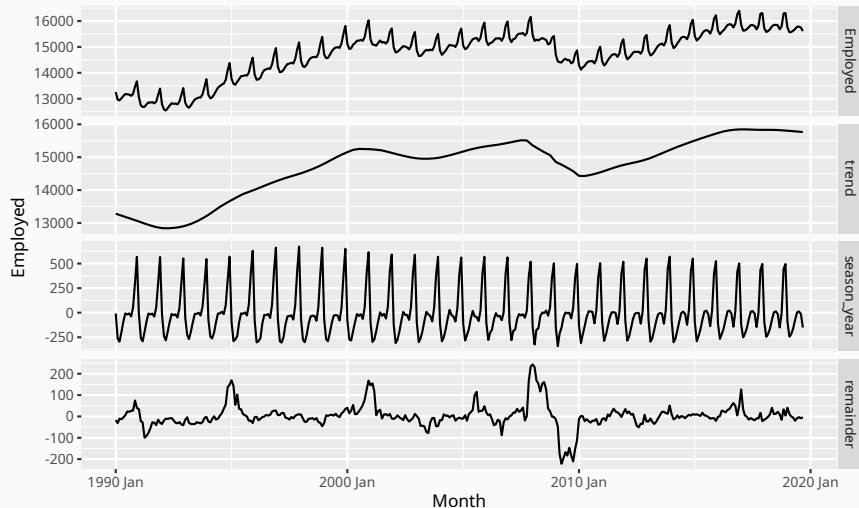
Seasonal adjustment

- We use estimates of S based on past values to seasonally adjust a current value.
- Seasonally adjusted series reflect **remainders** as well as **trend**. Therefore they are not “smooth” and “downturns” or “upturns” can be misleading.
- It is better to use the trend-cycle component to look for turning points.

STL decomposition

STL decomposition of US retail employment

season(window=05)



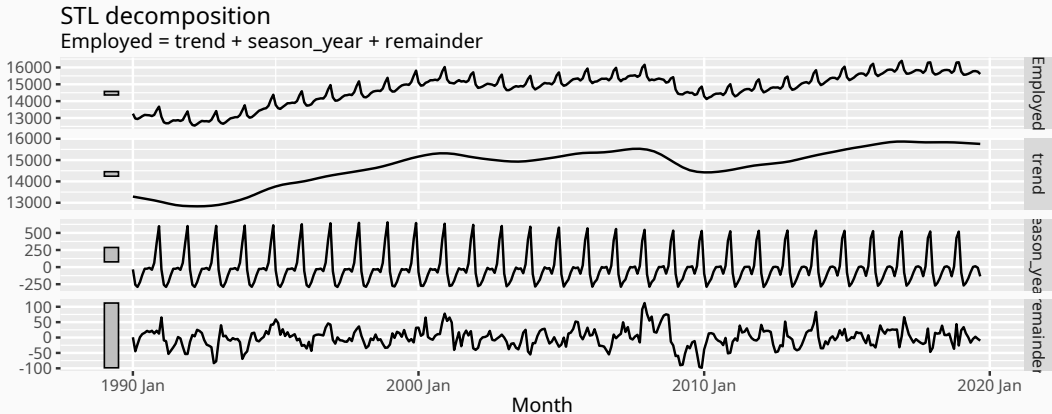
STL decomposition

```
us_retail_employment |>  
  model(STL(Employed ~ trend(window = 15) + season(window = "periodic"),  
    robust = TRUE  
  )) |>  
  components()
```

- `trend(window = ?)` controls wiggleness of trend component.
- `season(window = ?)` controls variation on seasonal component.
- `season(window = 'periodic')` is equivalent to an infinite window.

STL decomposition

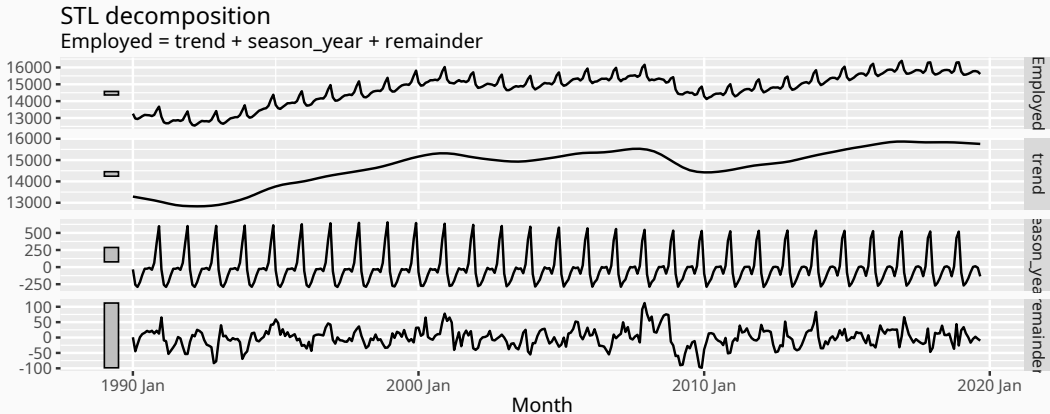
```
us_retail_employment |>  
  model(STL(Employed)) |>  
  components() |>  
  autoplot()
```



STL decomposition

```
us_retail_employment |>  
  model(STL(Employed)) |>  
  components() |>  
  autoplot()
```

- `STL()` chooses `season(window=13)` by default
- Can include transformations.



STL decomposition

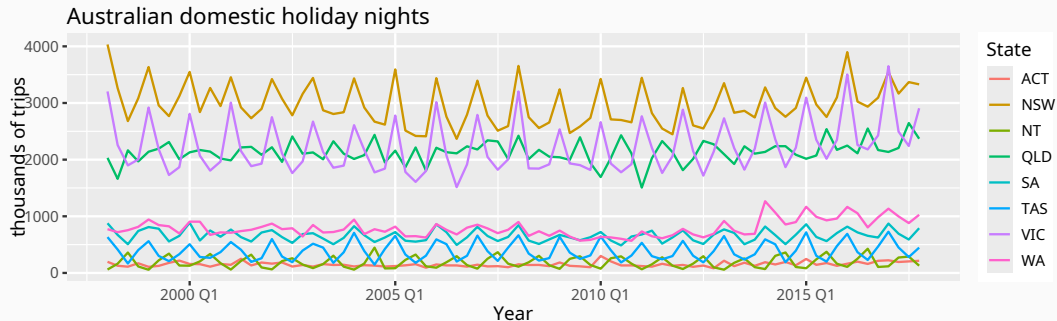
- Algorithm that updates trend and seasonal components iteratively.
- Starts with $\hat{T}_t = 0$
- Uses a mixture of loess and moving averages to successively refine the trend and seasonal estimates.
- trend window controls loess bandwidth on deasonalised values.
- season window controls loess bandwidth on detrended subseries.
- Robustness weights based on remainder.
- Default season: `window = 13`
- Default trend:

`window =`

`nextodd(ceiling((1.5*period)/(1-(1.5/s_season))))`

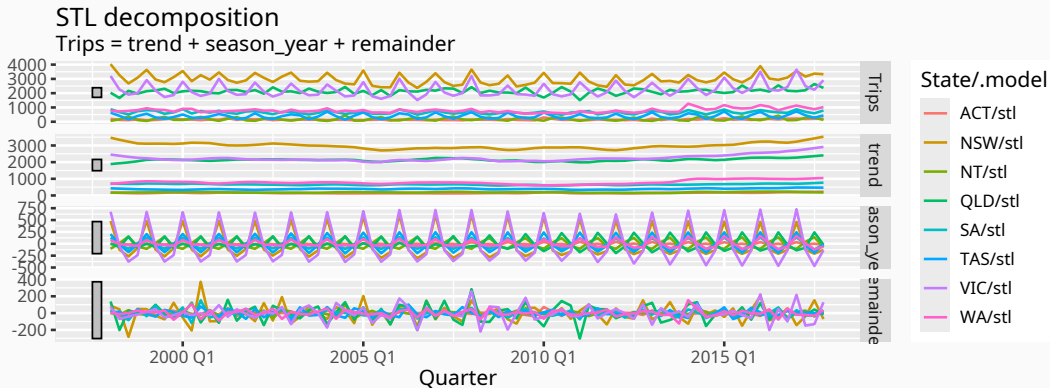
Australian holidays

```
holidays |> autoplot(Trips) +  
  labs(y = "thousands of trips", x = "Year",  
        title = "Australian domestic holiday nights")
```



Australian holidays

```
holidays |>  
  model(stl = STL(Trips)) |>  
  components() |>  
  autoplot()
```



Holidays decomposition

```
dcmp <- holidays |>
  model(stl = STL(Trips)) |>
  components()
dcmp
```

```
# A dable: 640 x 8 [1Q]
```

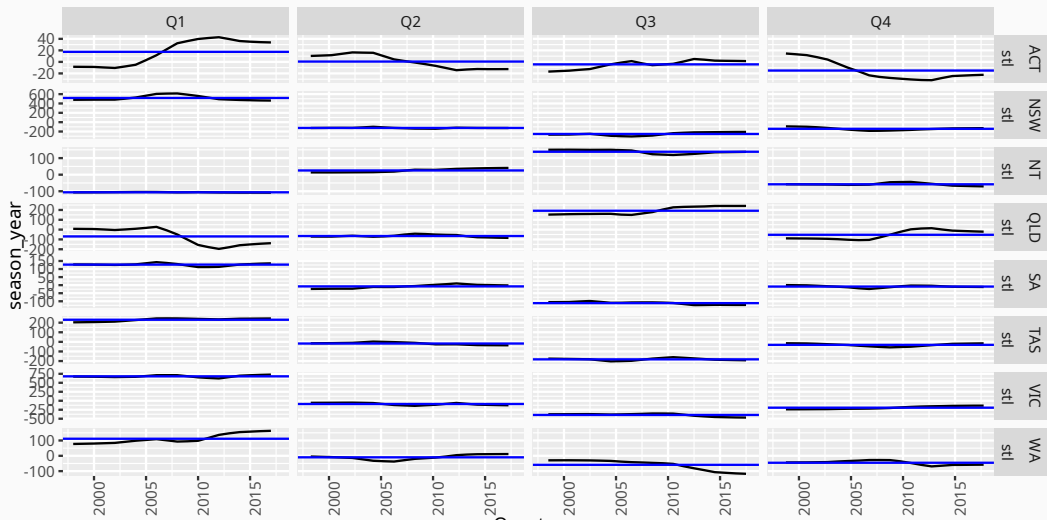
```
# Key:      State, .model [8]
```

```
# :        Trips = trend + season_year + remainder
```

	State	.model	Quarter	Trips	trend	season_year	remainder	season_adjust
	<chr>	<chr>	<qtr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	ACT	stl	1998 Q1	196.	172.	-8.48	32.6	205.
2	ACT	stl	1998 Q2	127.	157.	10.3	-40.6	116.
3	ACT	stl	1998 Q3	111.	142.	-16.8	-14.5	128.
4	ACT	stl	1998 Q4	170.	130.	14.6	25.6	156.
5	ACT	stl	1999 Q1	108.	135.	-8.63	-18.3	116.
6	ACT	stl	1999 Q2	125.	148.	11.0	-34.6	114.
7	ACT	stl	1999 Q3	178.	166.	-16.0	28.3	194.
8	ACT	stl	1999 Q4	218.	177.	13.2	27.5	204.
9	ACT	stl	2000 Q1	158.	169.	-8.75	-1.96	167.

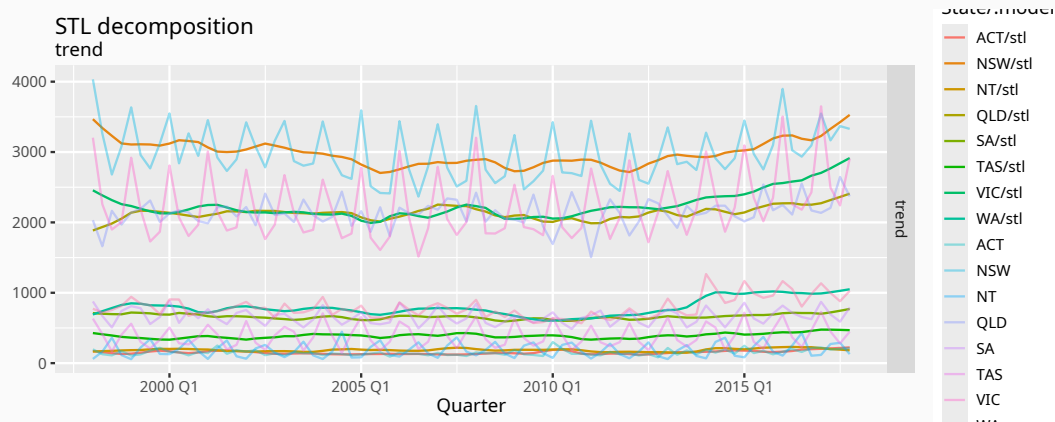
Holidays decomposition

```
dcmp |> gg_subseries(season_year)
```



Holidays decomposition

```
autoplot(dcmp, trend, scaleBars = FALSE) +  
  autolayer(holidays, alpha = 0.4)
```



Outline

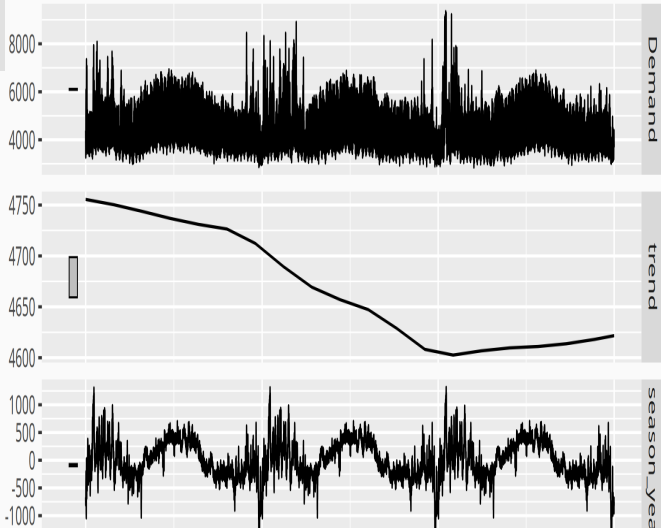
- 1 Transforming time series
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Multiple seasonality

```
vic_elec |>  
  model(STL(Demand)) |>  
  components() |>  
  autoplot()
```

STL decomposition

Demand = trend + season_year + season_week + season_day + season_hour +
remainder



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BREAKING NEWS

Police arrest man in connection with stabbing death of 17-year-old Masa Vukotic in M



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Treasurer Joe Hockey calls for answers over Australian Bureau of Statistics jobs data

By [Michael Vincent](#) and [Simon Frazer](#)

Updated 9 Oct 2014, 12:17pm

Federal Treasurer Joe Hockey says he wants answers to the problems the Australian Bureau of Statistics (ABS) has had with unemployment figures.

Mr Hockey, who is in the US to discuss Australia's G20 agenda, said last month's unemployment figures were "extraordinary".

The rate was 6.1 per cent after jumping to a 12-year high of 6.4 per cent the previous month.

The ABS has now taken the rare step of abandoning seasonal adjustment for its latest employment data.



PHOTO: Joe Hockey says he is unhappy with the volatility of ABS unemployment figures. (AAP: Alan Porritt)

RELATED STORY: [ABS abandons seasonal adjustment for latest jobs data](#)

The ABS stuff-up

BREAKING NEWS

Police arrest man in connection with stabbing death of 17-year-old Masa Vukotic in Mel

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ABS abandons seasonal adjustment for latest jobs data

By business reporter [Michael Janda](#)

Updated 8 Oct 2014, 4:19pm

The Australian Bureau of Statistics is taking the rare step of abandoning seasonal adjustment for its latest employment data.

The ABS uses seasonal adjustment, based on historical experience, to account for the normal variation between hiring and firing patterns between different months.

However, after a winter where the seasonally adjusted unemployment rate swung wildly from 6.1 to 6.4 and back to 6.1 per cent, [the bureau released a statement](#) saying it will not adjust the original figure for September for seasonal factors.

It will also reset the seasonal adjustment for July and August to one, meaning that these months will also reflect the original figures.

Sorry, this video has expired

VIDEO: Westpac chief economist Bill Evans discusses the ABS jobs data changes (ABC News)

RELATED STORY: Doubt the record breaking jobs figures? So does the ABS

RELATED STORY: Jobs increase record sees unemployment slashed

RELATED STORY: Unemployment surges to 12-year high at 6.4 pc

MAP: [Australia](#) 

The ABS stuff-up

ABS jobs and unemployment figures – key questions answered by an expert

A professor of statistics at Monash University explains exactly what is seasonal adjustment, why it matters and what went wrong in the July and August figures



📷 School leavers come on to the jobs market at the same time, causing a seasonal fluctuation. Photograph: Brian Snyder/Reuters

The Australian Bureau of Statistics has [retracted its seasonally adjusted employment data for July and August](#), which recorded huge swings in the jobless rate. The ABS is also planning to review the methods it uses for seasonal adjustment to ensure its figures are as accurate as possible. Rob Hyndman, a professor of statistics at Monash University and member of the bureau's methodology advisory board, answers our questions:

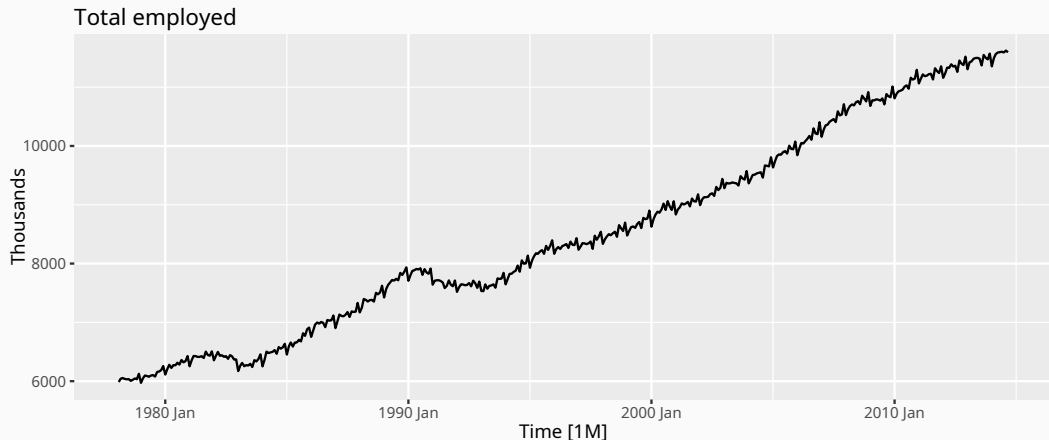
The ABS stuff-up

employed

```
# A tsibble: 440 x 4 [1M]
      Time Month  Year Employed
    <tmth> <ord> <dbl>    <dbl>
1 1978 Feb  Feb    1978    5986.
2 1978 Mar  Mar    1978    6041.
3 1978 Apr  Apr    1978    6054.
4 1978 May  May    1978    6038.
5 1978 Jun  Jun    1978    6031.
6 1978 Jul  Jul    1978    6036.
7 1978 Aug  Aug    1978    6005.
8 1978 Sep  Sep    1978    6024.
9 1978 Oct  Oct    1978    6046.
10 1978 Nov  Nov    1978    6034.
# i 430 more rows
```

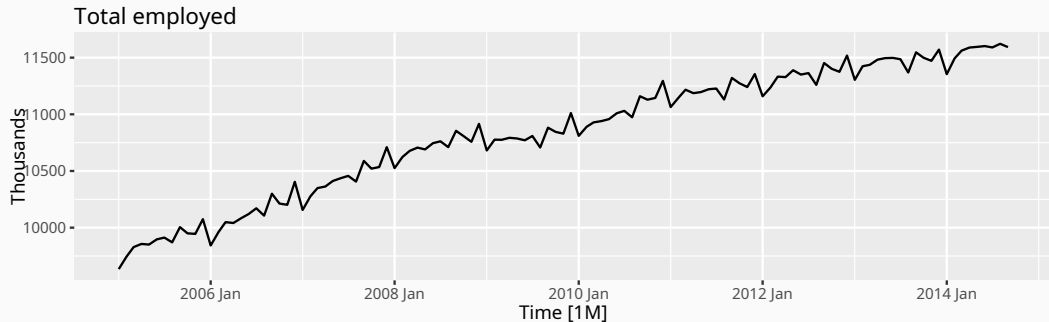
The ABS stuff-up

```
employed |>  
  autoplot(Employed) +  
  labs(title = "Total employed", y = "Thousands")
```



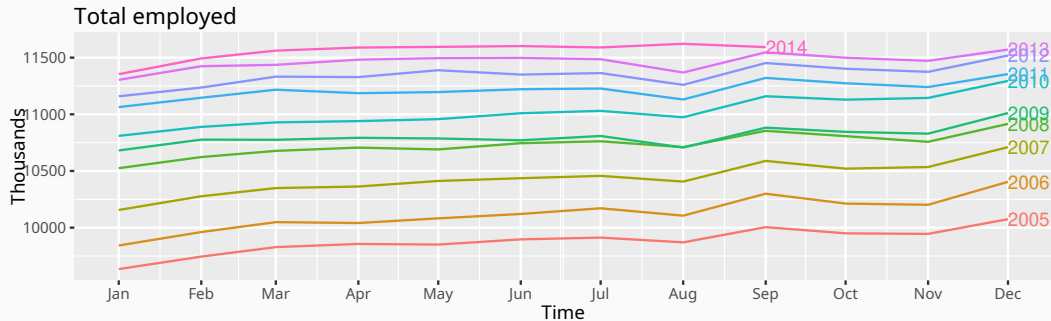
The ABS stuff-up

```
employed |>  
  filter(Year >= 2005) |>  
  autoplot(Employed) +  
  labs(title = "Total employed", y = "Thousands")
```



The ABS stuff-up

```
employed |>  
  filter(Year >= 2005) |>  
  gg_season(Employed, labels = "right") +  
  labs(title = "Total employed", y = "Thousands")
```



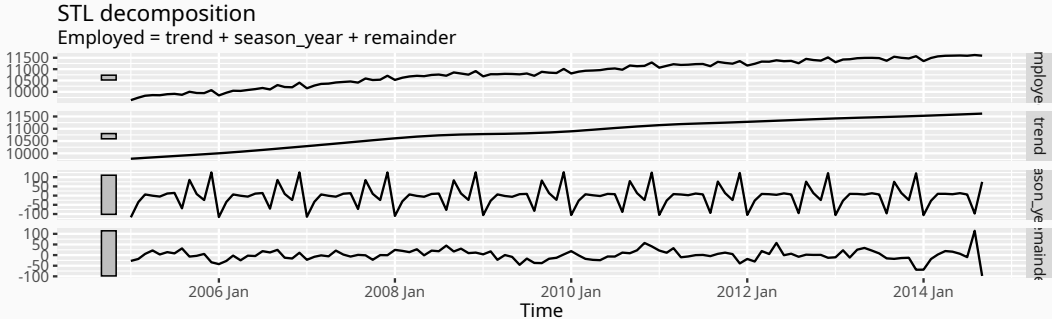
The ABS stuff-up

```
employed |>  
  mutate(diff = difference(Employed)) |>  
  filter(Month == "Sep") |>  
  ggplot(aes(y = diff, x = 1)) +  
  geom_boxplot() +  
  coord_flip() +  
  labs(title = "Sep - Aug: total employed", y = "Thousands") +  
  scale_x_continuous(breaks = NULL, labels = NULL)
```



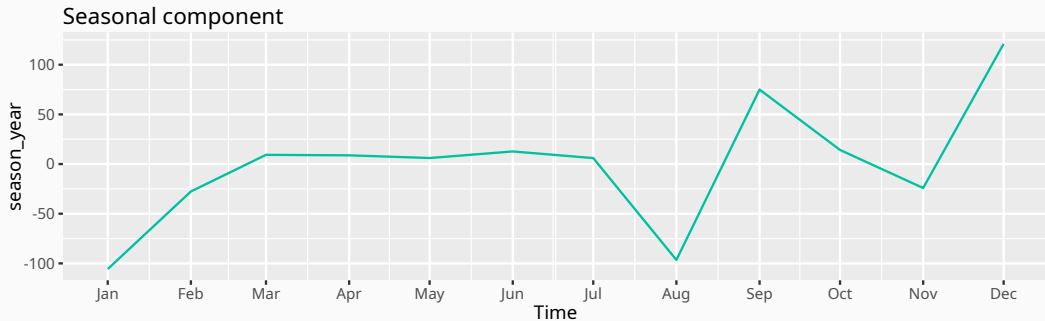
The ABS stuff-up

```
dcmp <- employed |>  
  filter(Year >= 2005) |>  
  model(stl = STL(Employed ~ season(window = 11), robust = TRUE))  
components(dcmp) |> autoplot()
```



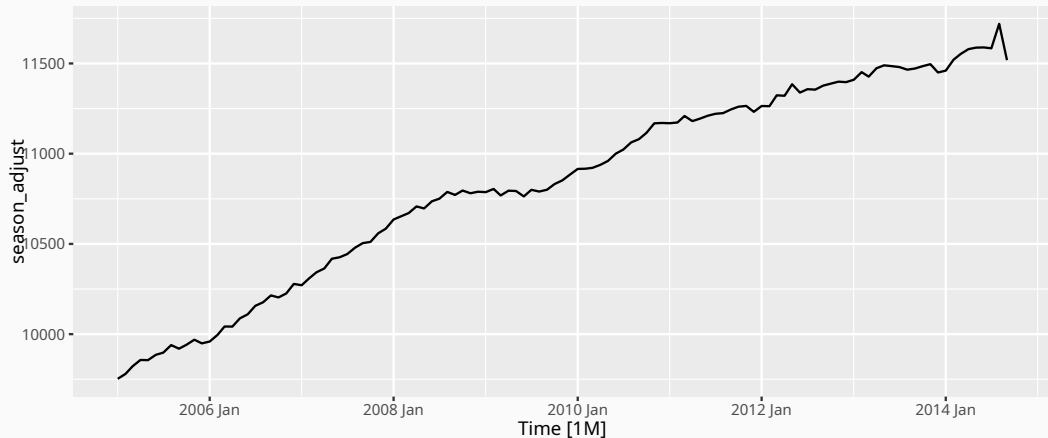
The ABS stuff-up

```
components(dcmp) |>  
  filter(year(Time) == 2013) |>  
  gg_season(season_year) +  
  labs(title = "Seasonal component") + guides(colour = "none")
```



The ABS stuff-up

```
components(dcmp) |>  
  as_tsibble() |>  
  autoplot(season_adjust)
```



The ABS stuff-up

- August 2014 employment numbers higher than expected.
- Supplementary survey usually conducted in August for employed people.
- Most likely, some employed people were claiming to be unemployed in August to avoid supplementary questions.
- Supplementary survey not run in 2014, so no motivation to lie about employment.
- In previous years, seasonal adjustment fixed the problem.
- The ABS has now adopted a new method to avoid the bias.